POWER-LAW MONTE CARLO SIMULATION

Comparing heavy-tailed return models and volatility dynamics for financial risk scenarios

Research question
Does combining GARCH-style volatility dynamics with power-law hybrid innovations improve probabilistic modelling of daily returns relative to simpler baselines?

Models compared
Normal IID; power-law hybrid IID; GARCH-normal; GARCH + power-law hybrid.

Assets represented in saved outputs
Reliance Industries, Titan Company, and Jio Financial Services. Jio has a shorter history than the other two.

Portfolio research report • 26 October 2026

1. Executive summary

The project compares four Monte Carlo return-model families and evaluates one-step-ahead predictive distributions using continuous ranked probability score (CRPS), negative log score (NLL), and 95% Value-at-Risk (VaR) exceedances.

Supported by saved outputs: hybrid GARCH has promising CRPS results in the reported within-training validation windows. On the previously inspected Titan test window, it has the lowest CRPS and NLL among the four models, and its VaR exceedance count is close to the 5% target. On Jio Financial's test window, the models are close: hybrid GARCH has slightly lower CRPS, while GARCH-normal has a slightly better NLL.

Not supported: the evidence does not establish universal model superiority, reliable stock-price prediction, trading alpha, or guaranteed loss reduction. Test windows were examined during development and must be described as exploratory.

2. Models and scope

Normal IID samples independent returns from a fitted normal distribution. Power-law hybrid IID uses an empirical body and Pareto-style tail. GARCH-normal uses GARCH(1,1)-style conditional variance with normal shocks. GARCH + power-law hybrid combines conditional volatility with heavy-tailed standardized innovations. Student's t innovations were excluded.

Saved outputs report 1,975 training and 494 test returns for Titan, 624 training and 157 test returns for Jio Financial, and a 494-day test horizon for Reliance. Gold is excluded from conclusions because the source series had a suspected adjustment/data anomaly.

The current test windows are not untouched final holdouts. The project is a risk/scenario-modelling prototype, not a price-target engine.

3. Selected simulation summaries

Values below come from archived model-summary CSVs. Daily returns and volatility are percentages; terminal prices are horizon-end simulated distributions. They are not freshly rerun during preparation of this report.

Titan Company

Model	Mean return/day	Volatility/day	VaR 95% loss	Median terminal price
Actual test period	0.06%	1.43%	1.90%	INR 4,381
Normal IID	0.11%	1.92%	3.05%	INR 5,774
Power-law hybrid IID	0.10%	2.18%	2.64%	INR 5,489
GARCH-normal	0.12%	1.87%	2.93%	INR 6,043
GARCH-power-law hybrid	0.12%	2.12%	2.31%	INR 5,928

Jio Financial Services

Model	Mean return/day	Volatility/day	VaR 95% loss	Median terminal price
Actual test period	-0.12%	1.90%	3.17%	INR 211
Normal IID	0.01%	2.08%	3.41%	INR 256
Power-law hybrid IID	0.01%	2.47%	2.97%	INR 257
GARCH-normal	-0.01%	2.20%	3.56%	INR 253
GARCH-power-law hybrid	-0.01%	2.33%	3.07%	INR 251

Reliance Industries

Model	Mean return/day	Volatility/day	VaR 95% loss	Median terminal price
Actual test period	-0.02%	1.32%	2.11%	INR 1,211
Normal IID	0.09%	1.78%	2.84%	INR 2,092
Power-law hybrid IID	0.09%	1.90%	2.38%	INR 2,070
GARCH-normal	0.09%	1.72%	2.72%	INR 2,069
GARCH-power-law hybrid	0.08%	1.72%	2.48%	INR 2,008

Interpretation: Reliance's realised terminal price was about INR 1,211, while the four simulated medians were about INR 2,008–2,092. This gap shows why a simulated median should not be marketed as an accurate price target.

4. Predictive-distribution evaluation

CRPS evaluates the overall predictive distribution; lower is better. NLL evaluates density assigned to the realised outcome; lower is better, even when scores are negative. VaR exception counts should be interpreted with sample size, uncertainty, and clustering in mind.

Titan Company — previously inspected test window

Model	Mean CRPS	Mean NLL	VaR exceptions	Expected at 5%
GARCH-normal	0.007667	-2.824849	18	24.70
GARCH-power-law hybrid	0.007579	-2.886925	25	24.70
Normal IID	0.007946	-2.766529	8	24.70
Power-law hybrid IID	0.007633	-2.862416	12	24.70

Jio Financial Services — previously inspected test window

Model	Mean CRPS	Mean NLL	VaR exceptions	Expected at 5%
GARCH-normal	0.010531	-2.554727	8	7.85
GARCH-power-law hybrid	0.010475	-2.552673	10	7.85
Normal IID	0.010605	-2.536297	7	7.85
Power-law hybrid IID	0.010577	-2.519665	11	7.85

Titan: hybrid GARCH has the lowest reported test CRPS (0.007579) and NLL (-2.886925), with 25 VaR exceptions versus 24.7 expected. Jio Financial: hybrid GARCH has slightly lower CRPS (0.010475) than GARCH-normal (0.010531), but GARCH-normal has a slightly better NLL (-2.554727 versus -2.552673). Jio's differences are small; no decisive winner is established.

5. Limitations and release requirements

Tail uncertainty: power-law estimates can be unstable when extreme observations are few, especially for the shorter Jio history. Threshold sensitivity should be reported.

Implementation: the GARCH code is custom-coded and should be described as GARCH-style until benchmarked against a specialist implementation.

Reproducibility: project files refer to different simulation counts, including 10,000 and 100,000. The final public release must state the exact count for each output. Do not imply that every chart displays 100,000 individual paths; some show summary bands.

Data rights: document data sources and adjustment rules; do not publish raw data unless redistribution is allowed.

Next steps: freeze the current version; standardise relative paths, seeds and dependencies; add unit tests; evaluate a fresh untouched holdout or pre-specified walk-forward protocol; quantify uncertainty; publish reproducible commands and limitations.

Conclusion

This is a useful research prototype demonstrating Python, Monte Carlo simulation, heavy-tail modelling, GARCH-style volatility dynamics, and probabilistic forecast evaluation. Its credibility comes from reproducibility and honest interpretation—not a claim that one model has definitively won.

Educational research only; not investment advice. All conclusions are conditional on the data, fitted models and saved experimental configuration.